

PropHedge

The Payoff Structure of Hedged Prop Firm Evaluation Accounts
with Static Drawdown: A Closed-Form Analysis

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Abstract

Proprietary trading firms (prop firms) sell access to evaluation accounts for a one-time fee. Profits on these accounts are partially paid out once the evaluation is passed, while losses are capped at the fee. This paper analyses the resulting payoff structure when the evaluation account is combined with an opposing, smaller position on a real brokerage account (live account). For evaluation accounts with a *static* drawdown, we model the combined position as a two-stage random walk with absorbing barriers, derive the PnL of every path and the expected value in closed form, and characterise the hedge ratios that maximise the Sortino ratio of the terminal-outcome distribution. Within the model, we find that: (i) there exist hedge ratios for which all terminal outcomes are positive; (ii) within the parameter range considered, the expected value is approximately linear in the maximum drawdown; (iii) the expected value is considerably more sensitive to the rule parameters of an evaluation than to its price; (iv) under a win rate of 50 %, daily drawdown limits and consistency rules are equivalent to a common upper bound on position size. We further quantify the sensitivity of the expected value to denied payouts, which constitute the dominant risk given that such combined positions typically violate the providers' terms of service.

Keywords: prop firm, evaluation account, payoff asymmetry, hedge ratio, gambler's ruin, Sortino ratio, expected value, static drawdown

Declaration of Generative AI Use: Claude (Anthropic) was used to assist in writing this paper.

1 Introduction

1.1 Prop Firms and Evaluation Accounts

Traditional proprietary trading firms trade with their own capital and give their traders a share of the profits. In recent years, a second business model aimed at retail traders has emerged: for a one-time fee, a trader purchases a *challenge* (also called an *evaluation*) on a simulated account with a nominal account size, for example \$50,000. If the trader reaches a predefined profit target without breaching a loss limit, they receive a so-called *funded account*. On this account, profits above a certain threshold are actually paid out at a fixed percentage (profit split).

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1.2 Rule Structure

A typical evaluation is characterised by the following rules:

Profit target Percentage gain relative to the account size that must be reached to pass the evaluation.

Maximum drawdown Loss limit that terminates the account immediately once reached. It can be defined as *static* (fixed relative to the starting capital) or *trailing* (moving relative to the account's high-water mark).

Daily drawdown Maximum permitted loss within a single trading day.

Consistency rule No single day may contribute more than a certain share of the total profit.

Payout conditions Profit threshold, profit split and minimum number of trading days required for a payout on the funded account.

The providers' business model relies on the fact that the vast majority of participants fail the evaluation and that the fees exceed the payouts to successful traders. For the participant, this results in a highly asymmetric payoff: the loss is capped at the fee, while the gain equals the funded account profit multiplied by the profit split. Structurally, this resembles a long option position whose premium is the evaluation fee.

1.3 Contribution

This paper studies the payoff of a combined position consisting of an evaluation account and an opposing position on a live account. The configuration is taken from the *Prop Arbitrage Simulator* by ATJ Research [1], which simulates it numerically across several phases. We translate it into a closed-form analytical model that yields the net PnL of every possible path, the expected value and a downside risk measure, and we characterise the hedge ratios that are optimal with respect to this measure. The analysis is descriptive: it examines the properties of the payoff structure created by prop firm rules and does not assess the feasibility or permissibility of such positions. Combining a prop account with an opposing personal position violates the terms of service of almost all providers; this is reflected in the model as the risk of denied payouts in Section 6.2.

2 The Combined Position

2.1 Definition

We consider a combined position in which every position on the prop account is accompanied by a position in the same instrument on a live account, in the *opposite direction* and of *smaller* size.

Definition 1 (Hedge ratio). Let X denote the position size on the prop account. The position size on the live account is

$$X^{\text{live}} = -h \cdot X, \quad h \in [0, 1].$$

The parameter h is called the *hedge ratio*. We write h_E for the evaluation phase and h_F for the funded phase.

If the prop account moves by an amount Δ , the live account moves by $-h\Delta$ (before fees). If the evaluation is failed, the live account gains a fraction h of the drawdown, which offsets the fee fully or partially. If the evaluation is passed, the live account loses a fraction h of the profit target, which is set against the potential payout in the funded phase.

2.2 Interior Hedge Ratios

A full hedge ($h = 1$) would neutralise every gain on the prop account with an equally large loss on the live account. Since the prop account carries no capital of its own, however, the two sides are not symmetric: the loss on the prop account is capped at the fee, while its gain is paid out only after passing and only partially. Hedge ratios that are optimal with respect to a risk-adjusted criterion therefore lie in the interior of the interval. In addition, the live account ties up real capital (Section 5.4).

2.3 Effect on the Outcome Distribution

Without the live position, the evaluation resembles a lottery: with high probability the fee is lost, with low probability a large payout follows. The opposing live position shifts value from the winning paths to the losing paths and thereby reshapes the distribution of terminal outcomes. Section 5 shows that for suitable (h_E, h_F) all terminal outcomes can become positive within the model.

3 Scope: Static Drawdown

This paper is restricted to evaluations with a *static* drawdown, where the loss limit is fixed at $A(1 - D)$ regardless of the account's history.

With a *trailing* drawdown, the loss limit moves up with the account's high-water mark. The remaining distance to the loss limit, and hence the probability of failure, then depends on the entire path so far. A constant hedge ratio is no longer optimal in this case; the optimal ratio becomes a function of the current state, which turns the problem into a dynamic control problem.

A state-dependent hedge ratio would moreover be subject to frictions that are difficult to capture in a closed-form model:

- **Execution costs:** Every adjustment of the live position incurs spreads, slippage and fees, which distort the effective hedge ratio.
- **Asynchrony:** The two accounts are held on different platforms, so that time lags between fills create unintended net exposure.
- **Granularity:** Contracts can only be traded in whole units (e.g. Micro E-mini S&P 500, MES). A continuously varying hedge ratio can therefore only be approximated, especially for small account sizes.

With a static drawdown, the geometry of the problem remains unchanged throughout each phase, so that one constant hedge ratio per phase is sufficient and the model remains tractable.

4 Model

4.1 Assumptions

1. Each trade wins or loses the fixed amount rA with probability $\frac{1}{2}$, where r is the fraction of the account size risked per trade.
2. The drawdown is static.
3. No other rule (daily drawdown, consistency, minimum trading days) is breached. Section 6.4 shows that this corresponds to an upper bound on r .

4. The live position is always held exactly at ratio h (no slippage).
5. The starting capital of the funded account is $A_F = A(1 + T_E)$; a fraction φ of the funded profit is paid out. In the numerical example, $\varphi = 1$.

4.2 Notation

Symbol	Meaning	Numerical example
A	Account size	\$50,000
C	Evaluation price	\$149
r	Fraction risked per trade	2 %
T_E, D_E	Profit target / max. drawdown, evaluation	8 % / 4 %
T_F, D_F	Payout target / max. drawdown, funded	8 % / 4 %
f_E	Fee per trade on the prop account	\$5
f_L	Fee per trade on the live account	\$2
φ	Profit split	100 %
h_E, h_F	Hedge ratio, evaluation / funded	optimised

Table 1: Model parameters and values used in the numerical example.

4.3 Transition Probabilities

Under Assumption 1, the account balance follows a symmetric random walk with step size rA that is absorbed at the upper barrier $+TA$ (pass) or the lower barrier $-DA$ (fail). By the classical gambler's ruin result [2, Ch. XIV.2], see also [4],

$$p_E = \frac{D_E}{D_E + T_E}, \quad p_F = \frac{D_F}{D_F + T_F}, \quad (1)$$

where p_E denotes the probability of passing the evaluation and p_F the conditional probability of reaching the payout target in the funded phase. These probabilities depend only on the ratio T/D and *not* on the position size r .

Combining both phases from (1) yields three terminal outcomes with unconditional probabilities

$$P_{FE} = 1 - p_E \quad (\text{evaluation failed}), \quad (2)$$

$$P_{PO} = p_E p_F \quad (\text{payout reached}), \quad (3)$$

$$P_{FF} = p_E (1 - p_F) \quad (\text{funded account lost}). \quad (4)$$

The probabilities are thus fully determined by the profit target and the maximum drawdown of the respective evaluation. For the parameters of the numerical example in Table 1 ($T = 8\%$, $D = 4\%$ in both phases), we obtain $p_E = p_F = \frac{1}{3}$, i.e. $P_{FE} = 66.7\%$, $P_{PO} = 11.1\%$ and $P_{FF} = 22.2\%$. An evaluation with $T = 10\%$ and $D = 5\%$ leads to the same result, whereas $T = 6\%$ and $D = 4\%$ yields $p = 40\%$.

4.4 Expected Number of Trades and Fees

For a symmetric random walk absorbed at $+a$ and $-b$ steps, the expected duration is $a \cdot b$ [2, Ch. XIV.3]. With $a = T/r$ and $b = D/r$, it follows that

$$n_E = \frac{T_E D_E}{r^2}, \quad n_F = \frac{T_F D_F}{r^2}. \quad (5)$$

The fees per phase are

$$F_E = n_E \left(f_E + f_L \mathbf{1}_{\{h_E > 0\}} \right), \quad F_F = n_F f_L, \quad (6)$$

with n_E and n_F from (5). Since the funded phase is only reached with probability p_E from (1), the expected total fees are

$$\mathbb{E}[F] = F_E + p_E F_F. \quad (7)$$

5 Path PnL and Expected Value

5.1 Composition of the PnL

The net PnL of a path consists of four components: the evaluation fee C , the PnL of the live position, the payout of the funded account, and the trading fees. If a phase ends at the drawdown, the live account gains hD ; if it ends at the target, it loses hT , each relative to the capital of that phase. The fees F_E and F_F are given by (6).

Evaluation passed (intermediate state).

$$\Pi_{PE} = -C - h_E T_E A - F_E. \quad (8)$$

Evaluation failed.

$$\Pi_{FE} = -C + h_E D_E A - F_E. \quad (9)$$

Funded account lost.

$$\Pi_{FF} = \Pi_{PE} + h_F D_F A_F - F_F. \quad (10)$$

Payout reached.

$$\Pi_{PO} = \Pi_{PE} - h_F T_F A_F + \varphi T_F A_F - F_F. \quad (11)$$

Here, Π_{PE} in (10) and (11) is the intermediate state from (8).

Figure 1 shows the paths (8)–(11) as a decision tree, evaluated at the Sortino-optimal hedge ratios of the numerical example, $h_E = 0.25$ and $h_F = 0.65$.

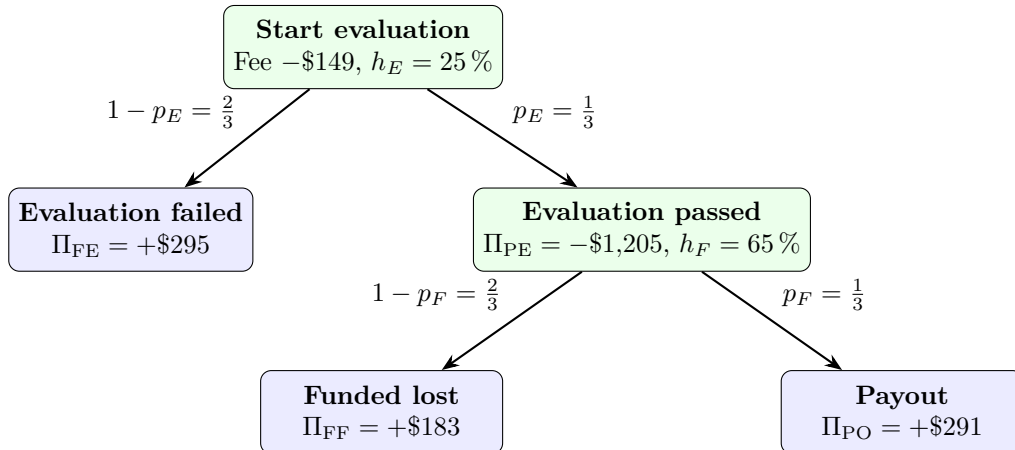


Figure 1: Decision tree of the combined position in the numerical example with $h_E = 0.25$, $h_F = 0.65$. Green: intermediate states, blue: terminal states.

5.2 Expected Value

The expected value is the sum of the terminal outcomes (9), (10) and (11), weighted by their probabilities (2)–(4):

$$\mathbb{E}[\Pi] = P_{\text{FE}} \Pi_{\text{FE}} + P_{\text{PO}} \Pi_{\text{PO}} + P_{\text{FF}} \Pi_{\text{FF}}. \quad (12)$$

In the numerical example, this gives

$$\mathbb{E}[\Pi] = \frac{2}{3} \cdot 295 + \frac{1}{9} \cdot 291 + \frac{2}{9} \cdot 183 \approx \$269.67.$$

5.3 Risk Measure and Optimal Hedge Ratios

To characterise the hedge ratios, we use the Sortino ratio [3]

$$S(h_E, h_F) = \frac{\mathbb{E}[\Pi]}{\sigma_-}, \quad \sigma_-^2 = \sum_{i \in \{\text{FE}, \text{PO}, \text{FF}\}} P_i \min(\Pi_i - \mathbb{E}[\Pi], 0)^2, \quad (13)$$

with $\mathbb{E}[\Pi]$ from (12).

Unlike its usual application, the Sortino ratio here does not refer to a time series of returns but to the distribution of the three terminal outcomes Π_{FE} , Π_{PO} and Π_{FF} . The downside deviation σ_- measures how far the terminal outcomes fall below the expected value, weighted by their probabilities. Maximising S in (13) therefore identifies the hedge ratios for which the outcomes of the individual paths are most concentrated relative to their expected value.

The maximiser is determined by grid search over $h_E, h_F \in \{0, 0.05, \dots, 1\}$. Figure 2 shows the result for the numerical example. The Sortino ratio is sharply concentrated around the optimum at $h_E = 0.25$ and $h_F = 0.65$, where it reaches a value of approximately 6.6. At this point, all three terminal outcomes lie between \$183 and \$295 (Figure 1). Small deviations from the optimum, particularly in h_E , reduce the Sortino ratio considerably. For larger h_E , the loss on the passing path cannot be compensated in the funded phase; for smaller h_E , the fee is not offset on the failing path.

5.4 Capital Requirement

The prop account is simulated, whereas the live account carries real capital. The lowest running PnL occurs on the payout path immediately before the payout, when both live losses have been realised but the payout has not yet been received:

$$\Pi_{\min} = \Pi_{\text{PE}} - h_F T_F A_F - F_F, \quad (14)$$

which corresponds to (11) without the payout term $\varphi T_F A_F$. In the numerical example, $\Pi_{\min} = -\$4,029$. Together with the margin for the live position, this value determines the capital tied up by the combined position.

6 Results

6.1 Expected Value and Maximum Drawdown

Table 2 shows the expected value according to (12) with $h_E = 0.25$ and $h_F = 0.65$ for different maximum drawdowns, all other parameters unchanged. If the drawdown is increased equally in both phases ($D_E = D_F = D$), the expected value increases approximately linearly by about \$130–\$140 per percentage point in the range from 2% to 8%.

A larger drawdown increases the pass probabilities p_E and p_F and thus the expected payout. At the same time, the live account gains hDA on the failing path, so that the fee is offset at smaller hedge ratios. This is counteracted by the trading fees, which grow linearly with D

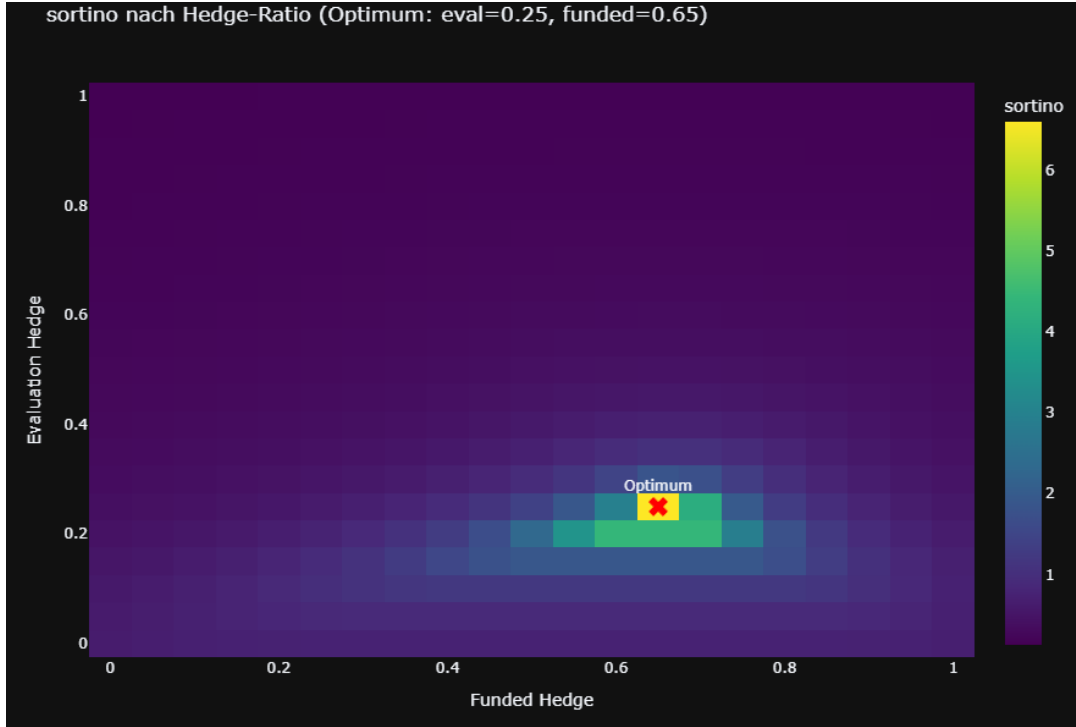


Figure 2: Sortino ratio of the terminal outcomes as a function of the hedge ratios h_E (vertical) and h_F (horizontal) for the numerical example. The optimum lies at $h_E = 0.25$, $h_F = 0.65$.

according to (5). Strictly speaking, $\mathbb{E}[\Pi]$ is not linear in D ; however, the quadratic dependence on $p_E p_F$ is offset by the saturation of $D/(D+T)$ in such a way that an almost linear relationship emerges within the parameter range considered (Figure 3). If only the evaluation drawdown is increased, the increase flattens considerably.

6.2 Sensitivity to Denied Payouts

Since most prop firms prohibit opposing positions on personal accounts, denied payouts must be taken into account. Let $q \in [0, 1]$ be the probability that a reached payout is actually paid. The terminal state “payout” then splits: with probability q the outcome is Π_{PO} , with probability $1 - q$ it is

$$\Pi_{PO}^{\text{denied}} = \Pi_{PE} - h_F T_F A_F - F_F = \Pi_{\min},$$

with $\Pi_{\min}$ from (14), since the live loss of the funded phase has already been realised when the payout is rejected. The two variants of the payout state differ by exactly the payout $\varphi T_F A_F$. Hence

$$\mathbb{E}[\Pi](q) = \mathbb{E}[\Pi] - (1 - q) p_E p_F \varphi T_F A_F, \quad (15)$$

where $\mathbb{E}[\Pi]$ is the expected value from (12). Setting (15) to zero gives the breakeven payout rate

$$q^* = 1 - \frac{\mathbb{E}[\Pi]}{p_E p_F \varphi T_F A_F}. \quad (16)$$

In the numerical example with $h_E = 0.25$ and $h_F = 0.65$, (16) yields $q^* = 1 - 269.67/480 \approx 43.8\%$. The expected value of the combined position is therefore positive only if more than about 44% of reached payouts are honoured.

At the Sortino-optimal hedge ratios, the most likely terminal state, failing the evaluation ($P_{FE} = \frac{2}{3}$), yields +\$295 independently of any payout. The dependence on q is therefore concentrated entirely on the payout path, which also carries the lowest running PnL $\Pi_{\min}$.

D	p_E	$\mathbb{E}[\Pi], D_E = D_F = D$	$\mathbb{E}[\Pi], \text{ only } D_E = D$
2 %	20.0 %	−\$5.8	\$107.8
3 %	27.3 %	\$127.0	\$197.4
4 %	33.3 %	\$269.7	\$269.7
5 %	38.5 %	\$412.4	\$328.7
6 %	42.9 %	\$550.2	\$377.3
7 %	46.7 %	\$680.7	\$417.5
8 %	50.0 %	\$803.0	\$451.0

Table 2: Expected value as a function of the maximum drawdown (other parameters as in the numerical example).

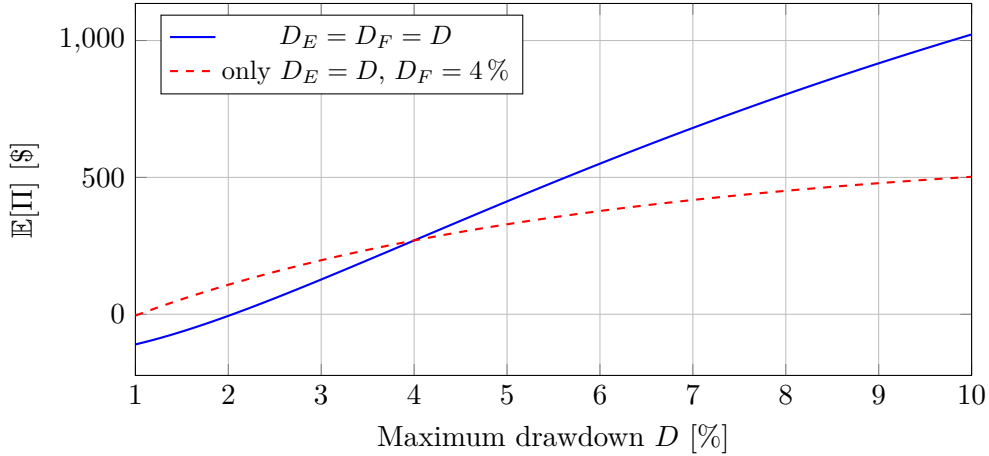


Figure 3: Expected value as a function of the maximum drawdown according to Equation (12) with $h_E = 0.25$, $h_F = 0.65$.

6.3 Sensitivity to Price and Rules

Since the fee C enters every terminal outcome with a factor of -1 , $\partial \mathbb{E}[\Pi] / \partial C = -1$: every dollar of fee reduces the expected value by exactly one dollar. The rules, in contrast, act multiplicatively through the pass probabilities on the comparatively large payout $\varphi T_F A_F$. Table 3 compares both effects.

In the numerical example, a single additional percentage point of drawdown changes the expected value by almost as much as the entire fee. The valuation of an evaluation within the model is therefore dominated by its rule parameters (drawdown, profit target, profit split) rather than by its price. This mirrors the option analogy from the introduction: the fee acts as an additive premium, whereas the rules determine the distribution of the underlying payoff.

6.4 Daily Drawdown and Consistency Rule as a Position Size Bound

According to (1), with a win rate of exactly 50 % the pass probabilities do not depend on the position size r . If at most k trades are executed per day, the two remaining rules translate as follows:

- **Daily drawdown** with limit d : the maximum daily loss is krA . The rule is satisfied if $kr \leq d$.
- **Consistency rule** with share c : the maximum daily profit is krA and may not exceed cTA . The rule is satisfied if $kr \leq cT$.

Change relative to numerical example	$\mathbb{E}[\Pi]$	$\Delta\mathbb{E}[\Pi]$
Numerical example	\$269.7	–
Price C : \$149 $\rightarrow$ \$99	\$319.7	+\$50.0
Price C : \$149 $\rightarrow$ \$299	\$119.7	–\$150.0
Drawdown $D_E = D_F$: 4 % $\rightarrow$ 5 %	\$412.4	+\$142.7
Drawdown $D_E = D_F$: 4 % $\rightarrow$ 3 %	\$127.0	–\$142.7
Profit target T_E : 8 % $\rightarrow$ 6 %	\$367.9	+\$98.2
Profit target T_E : 8 % $\rightarrow$ 10 %	\$195.5	–\$74.2
Profit split φ : 100 % $\rightarrow$ 80 %	\$173.7	–\$96.0

Table 3: Sensitivity of the expected value to price and rules.

Both rules thus reduce to the same type of constraint, namely an upper bound on the position size:

$$r \leq r_{\max} = \frac{1}{k} \min(d, cT). \quad (17)$$

Within this bound, the rules do not affect the distribution of terminal outcomes. They affect the expected value only through the number of trades, which grows with $1/r^2$ according to (5), and through the duration of the evaluation. Table 4 illustrates this effect.

r	n_E	n_F	$\mathbb{E}[\Pi]$
0.5 %	128	128	–\$650.3
1 %	32	32	\$85.7
2 %	8	8	\$269.7
4 %	2	2	\$315.7

Table 4: Effect of position size on the number of trades and the expected value (numerical example, $h_E > 0$).

Since $\mathbb{E}[\Pi]$ is increasing in r through the fee term alone, it attains its maximum at the boundary $r = r_{\max}$ from (17). A daily drawdown or consistency rule is therefore costly within the model exactly when it lowers $r_{\max}$ so far that the fees consume the payout term. This allows both rules to be compared on a common scale via their implied bound $r_{\max}$, which in turn depends inversely on the number of trades per day k .

7 Limitations

- A win rate of exactly 50 % and symmetric gains and losses are an idealisation; spread and slippage shift the live position slightly into negative expected value.
- The analysis in Section 6.2 shows that the providers' prohibition of such positions is the central risk. The rate q is unknown and provider-dependent.
- The assumption $A_F = A(1 + T_E)$ deviates from providers that reset the funded account to the original account size. Under this assumption, the funded account is larger than the original account by the factor $(1 + T_E)$, so that a given h_F implies proportionally larger live positions and thus more capital and margin. In the numerical example, $|\Pi_{\min}|$ from (14) is about \$208 larger than with a funded account reset to A .

- Only a single payout is considered. Multiple consecutive payouts on the same funded account would increase the expected value.
- Contract granularity and margin requirements are only approximately accounted for.
- Trailing drawdowns are excluded; they lead to a dynamic control problem that is left for future research.

8 Conclusion

For evaluation accounts with a static drawdown, the payoff of an evaluation combined with an opposing live position can be described fully in closed form. The expected value follows from the PnL of the three possible paths, weighted by their probabilities. The live position reshapes the outcome distribution such that, for suitable hedge ratios, all terminal outcomes are positive within the model. Within the parameter range considered, the expected value is approximately linear in the maximum drawdown, and it is considerably more sensitive to the rule parameters than to the price. Under a win rate of 50 %, daily drawdown and consistency rules are equivalent to an upper bound on position size. The dominant risk is the denial of payouts: in the numerical example, the expected value is positive only if at least about 44 % of reached payouts are honoured.

Disclaimer

This paper is a purely analytical study of a payoff structure. It does not constitute investment, financial or legal advice and does not recommend or encourage any trading activity or the violation of the terms of service of prop firms or brokers. Holding opposing positions on a prop account and a personal account is prohibited by most providers and may result in account termination and denied payouts. Trading futures and other financial instruments involves substantial risk of loss.

Code Availability

The Python code for computing the path PnL, the expected value and the Sortino-optimal hedge ratios is publicly available at <https://github.com/Lennart967/PropHedge>.

References

- [1] ATJ Research. *Prop Arbitrage Simulator*. Online tool, <https://proparbitragesimulator.atjresearch.com/>, accessed October 2026.
- [2] W. Feller. *An Introduction to Probability Theory and Its Applications*, Vol. I, 3rd ed. Wiley, New York, 1968.
- [3] F. A. Sortino and L. N. Price. Performance Measurement in a Downside Risk Framework. *The Journal of Investing*, 3(3):59–64, 1994.
- [4] D. Williams. *Probability with Martingales*. Cambridge University Press, Cambridge, 1991.